

Integration of Battery Swapping Station into Microgrid: A Review

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Abstract—Electric Vehicles (EVs) are considered a prominent alternative to fossil fuel-based vehicles to reduce environmental pollution in the transportation sector. Charging infrastructure development, especially fast-charging stations, is quite excellent in terms of charging time and speed compared to the other charging options. But that will reduce the battery lifetime. On the other hand, Battery Swapping Station (BSS) will swap batteries within ten minutes. As here, there is no need for fast charging of batteries; it will increase the lifetime. This paper presents a detailed and systematic review of BSS integration into the power system. Also, the concept of BSS-Microgrid is presented where the BSS can act as an Energy Storage System (ESS) upon requirement. The various optimization modeling solution techniques implemented in the literature and the challenges are discussed thoroughly.

Keywords—Battery Swapping Station, Electric Vehicle, Microgrid, Optimization, Charging Infrastructure

I. INTRODUCTION

Air pollution, alongside climate change, is the biggest threat to the upcoming generation regarding human health. Countries like China, the USA, India, and Russia contribute to carbon dioxide emissions. To achieve the net-zero emission target use of electric vehicles in transportation proves to be a great solution. Also, it will phase out the use of fossil fuel-based vehicles. In the last decade, the adoption of electric vehicles increased rapidly. Much research is done in the literature on EV charging infrastructure developments.

Despite many advancements in the EV charging area, high charging time is still a headache. Unlike ICE-based fuel stations, where it takes around 5 to 10 minutes to fill the fuel, an EV charging station takes around one to three hours to charge the vehicle, depending upon the charging station. So, people are moving towards a new idea called Battery Swapping Stations (BSS), which will replace the discharged or used batteries with a new set of fully charged ones.

Moreover, BSS can take part in the DR programs to take power during off-peak hours to charge the reserved batteries and act as a battery energy storage source (BESS) during peak hours in a microgrid environment. Hence it can save the charging time for the incoming electric vehicles and reduce the stress on the electrical network by acting as BESS. There has been a massive debate on choosing the technology between EV fast-charging stations and BSS in the recent past.

The vehicle battery size, the maximum speed of the vehicle, the fast-charging station power requirement, and overall BSS operational cost are extensively discussed. In [1], a Monte Carlo simulation approach is adopted to get the

vehicle owner behaviors for comparing the earning and service capability of both EV fast charging and BSS. Their results show that BSS gives more financial and social benefits over fast-charging stations with similar service capabilities. So, the main benefits of the BSS are listed as follows.

- Unlike at fast-charging stations, vehicle batteries should be charged faster to fulfill customer satisfaction.
- In BSS, the depleted batteries can be charged slowly and increase battery life.
- The depleted batteries can be charged during off-peak hours by participating in the DR program, limiting the power system infrastructure up-gradation, and reducing related costs.
- It can act as a BESS for the local microgrid when required.
- BSS can take power from renewable energy resources like solar PV and wind energy equipped with advanced controllers to charge the discharged battery.
- BSS can be combined with existing fast-charging stations to extract the combined benefits. It is only possible by having the most advanced optimal control strategy.

Even though there are not many BSSs deployed in the past, the pilot projects show outstanding results across the globe, especially in China, in terms of providing excellent service and making a profit. The main contributions of this paper are as follows. It provides a well-analyzed and detailed review of the BSS and BSS-Microgrid architecture. We highlight and discuss various optimization models and solution techniques used for both cases. At last, we highlight various challenges associated with the BSS implementation.

The rest of the paper is organized as follows. Section II briefly discussed the battery swapping system. In section III the problems associated with the BSS implementation are discussed. Section IV reviews optimization modeling techniques applied in the literature for both the BSS and BSS-Microgrid models. In section V, a typical BSS-Microgrid uncertainty modeling is presented. Section VI discussed the challenges associated with the BSS implementation. Lastly, section VII concludes the paper.

II. BATTERY SWAPPING SYSTEM

As discussed earlier, BSS can replace its charged battery with a discharged one and help the local microgrid with critical aspects like load balancing, peak shaving, etc., by acting as an energy storage system (ESS). In this section, we have discussed some of the main aspects of BSS, starting from its general architecture to its integration into a local microgrid. We have also discussed various pilot projects going on worldwide on BSS.

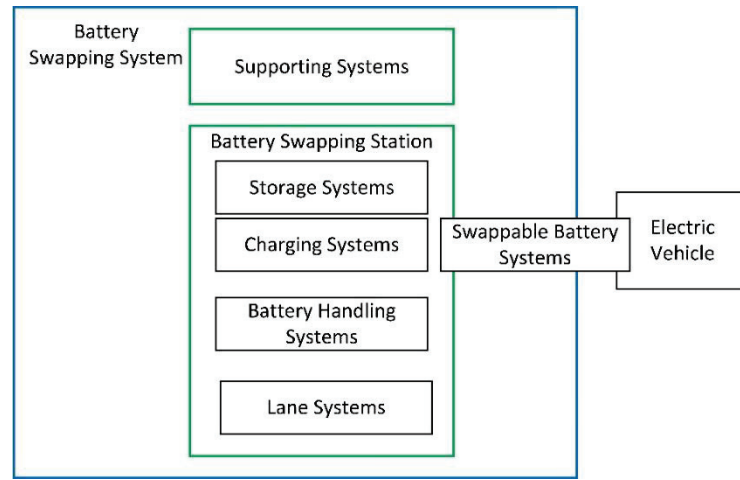


Fig. 1. General layout of Battery Swapping System

Fig. 1 shows the general architecture of a Battery Swapping System. The battery-swapping station, which consists of the storage system, charging system, battery handling system, and lane system, is the main element of a battery swapping system. An EV will come to the BSS, and it will interact with the swappable battery system and exchange its depleted battery with a fully charged one [1]. In [1], a detailed comparison between the available charging technology (AC and DC) and BSS in terms of time at the station, charging speed, battery life, various costs, and standards is discussed.

Table 1 compares the AC charging station, DC charging station, and BSS. The EV waiting time at the BSS station is far less than both AC and DC charging stations. Other advantages include charging speed and battery life. As of now, due to limited advancement in the EV battery technology, BSS batteries need to be purchased during investment time, so investment cost is highest compared to charging stations. But by seeing the recent trend in the advancement in battery technology, both size and capacity of the battery will improve manifold in the future. With that, the investment cost will subsequently reduce. In literature broadly, two types of research have been done on BSS. One category is optimal siting, cost-benefit analysis, and BSS analysis during the planning stage. The other one is about the utilization of BSS as an energy storage source in a microgrid environment in association with various renewable energy sources.

III. PROBLEMS DISCUSSED

This review paper briefly discusses the problems and solution approaches associated with BSS from the planning stages to the implementation stage. Problems during the planning stages include the optimal siting and sizing of BSS, optimal cost-benefit analysis, feasibility analysis, and battery ownership. Other problems related to BSS implementation and running stages include operational cost minimization, locally available renewable sources integration and maximum utilization, and operating BSS as energy storage for the local microgrid when required other than battery swapping. Significant challenges are elaborately discussed in section VI. A review of some of the recent literature where extensive analyses are done on the above-discussed problems along with proposed solution approaches and results are presented in section IV.

IV. LITERATURE REVIEW OF BSS AND BSS-MICRO GRID

In [1], the Monte Carlo simulation method compares BSS properties with EV charging stations in terms of various service capacities. The authors have proved that more social and financial benefits can be achieved with BSS than with charging stations with the same capacity. In [2], a generalized benders decomposition technique is used to minimize the charging cost fulfilling EV demand, and the simulations are done on real-world data. Similarly, in [3], the authors have worked on battery standardization and technologies and deployment of BSS during the planning stage.

TABLE I. Comparison of BSS with fast charging stations.

	AC Charging Station	DC Charging Station	BSS
Time at station	Long	Short to medium	Fixed, Short
Charging speed	1.4-20 kW	36-90 kW	0.3 C
Battery life	Long	Medium to short	Long
Investment cost	Low	Medium to high	High
Operational cost	Low	Medium to low	Medium to high
Additional battery	No	No to low	Low to high
Standard	J1772, IEC 62192 Type-2, GB/T 20234	J1772 Combo, CHAdeMO, GB/T 27930	IEC 62840, GB/T 29772

The authors in [4] selected BSS locations using massive GPS data e-taxis. They used the hidden Markov model and clustering technique and calculated the mean distance error between BSS generated demand and BSS location. The metric is reduced and proved to be reducing the range anxiety problem. In [5], [6], various algorithms use multi-objective optimization techniques to optimize cost and economic aspects. In [6], a PV-based BSS system verifies the day-ahead scheduling problem to minimize purchased electricity costs.

Similarly, in [7], a PV-based BSS is proposed to maximize the utilization of PV energy. In [8], the authors have analyzed an optimal cost-benefit using a population-based algorithm and verified it by two different IEEE bus systems. In [9], [10], authors have worked on optimal routing and scheduling of BSS considering e-buses for the travel range prediction. Table 2 shows the detailed analysis of various approaches to adopting BSS and its impact on the power system.

Now coming to the second aspect of BSS and that is fulfilling EV charging demand, it can be helpful to the local microgrid. So many debates were there on choosing between BSS and charging station (CS) in the past. One of the problems that came to notice is that if a BSS is established, there should be an adequate number of batteries available at the station, which will add more cost. But if those batteries can be used as ESS and participate in the demand response program, that will be a significant advantage. In [11], authors have investigated the optimal location of BSS in a local microgrid (MG) and getting complete services from the BSS. A mixed-

integer nonlinear programming problem is formulated, and BSS-MG coordination is examined.

Similarly, in [12], authors have proposed multiple shadow-based BSS-MG coordination methods, eliminating various barriers to the practical coordination among them. In [13], the active participation of BSS in the economic operation of an isolated MG is discussed. In [14], authors have investigated BSS as a cumulative source to collect the number of fully charged batteries from different nanogrids and charge the EVs.

The fully charged batteries will be charged by various distributed energy resources present in the nanogrids, and they are transported to the BSS via trucks. In [15], how BSS can be utilized as an ESS to support microgrid is studied, and the overall problem is formulated as mixed-integer linear programming.

V. TYPICAL BSS-MICROGRID UNCERTAINTY MODELLING

As discussed previously, when the microgrid gets disconnected from the main grid during any emergency condition, the BSS battery bank can fulfill the load requirement and solar and wind generation. Fig. 2 shows a typical layout of the BSS-microgrid model. As we can see, BSS can exchange its battery energy with a microgrid in two different ways. It can charge its battery pack by taking energy from the microgrid during off-peak hours in the microgrid to BSS (M2B) mode. Also, it can act as an additional energy

TABLE II. Modeling objectives and solution techniques used for BSS and BSS-Microgrid.

Reference	Objective	Method used	Remarks
[1]	Comparison with EV charging station done in terms of various service capacities	Monte Carlo simulation	More social and financial benefits than CS with the same capacity.
[2]	Charging cost minimization satisfying EV demand	Generalized benders decomposition	Simulations were done on real-time data.
[3]	Help to deploy BSS in the planning stage	Robust optimization	Battery standardization and technologies are discussed.
[4]	BSS location selection using a large scale of GPS data of taxis	Hidden Markov model, clustering technique	Mean distance error reduction by 52.5% and 62.7%.
[5]	Maximization of charging service capacity and minimization of the total cost	Covariance matrix adaption evaluation strategy-based algorithm	Peak demand reduced; total cost reduced by 12 %.
[6]	Day-ahead economic scheduling to minimize purchased electricity cost from grid	Chance-constraint programming and genetic algorithm	A PV-based BSS system is used for verification.
[8]	Optimal cost-benefit analysis	Modified differential evolution algorithm	Verified using modified IEEE 15-bus and 43-bus radial system.
[10]	Optimal location of BSS for e-Buses in a traffic network of South Korea	Set-covering-based model, flow-based model, and path-based model	e-Buses travel range, minimizing number of BSS are considered.
[9]	Routing and schedule planning for battery swapping system	A novel algorithm to calculate optimal scales of battery swap station	MATLAB simulations were done based on Xuejiadao BSS having six e-Buses in China.
[7]	A PV-based BSS is proposed with maximum utilization of PV energy	Power distribution model and battery swap service model are developed	-

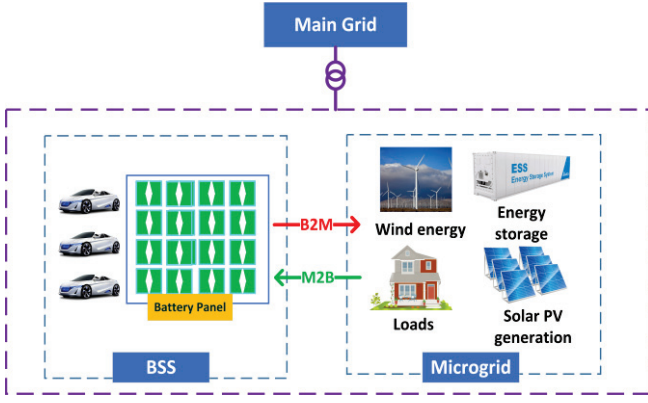


Fig. 2. Typical layout of BSS-Microgrid model.

storage system during peak hours and supply energy back to the microgrid in BSS to microgrid (B2M) mode. But solar and wind generations are coming under the non-dispatchable energy sources category and have uncertainties. In the case of BSS, the electric vehicle arrival time and arrival rate also possess uncertainty. So proper modeling of the BSS-microgrid components considering uncertainty is required to analyze the system. This part presents typical uncertainty models of the BSS-microgrid components like solar PV, wind power output, and EV arrival time.

A. Solar PV Power:

The PV power mainly depends upon the solar irradiance level, and the relation between them is stated as [16]

$$P^{PV} = \chi A^{PV} \eta^{PV} \quad (1)$$

Where χ represents the solar irradiance level, A^{PV} represents the solar radiation area, and η^{PV} represents the efficiency. From equation (1), the PV power is directly proportional to the solar irradiance level. The literature demonstrates that a day's per hour solar irradiance follows the famous Beta distribution [17]. The corresponding probability distribution function for the solar irradiance level is written as

$$SI(i) = \frac{\Gamma(\lambda_1) + \Gamma(\lambda_2)}{\Gamma(\lambda_1) * \Gamma(\lambda_2)} \left(\frac{i}{i_{\max}} \right)^{\lambda_1 - 1} \left(1 - \frac{i}{i_{\max}} \right)^{\lambda_2 - 1} \quad (2)$$

Where the variable i represents the light intensity and $i_{\max}$ is the maximum value of it. λ_1 and λ_2 are represents the shape factors. Similarly the PV power output can be formulated as Beta distribution PDF and can be written as

$$PS(P^{out}) = \frac{\Gamma(\lambda_1) + \Gamma(\lambda_2)}{\Gamma(\lambda_1) * \Gamma(\lambda_2)} \left(\frac{P^{out}}{P_{\max}^{out}} \right)^{\lambda_1 - 1} \left(1 - \frac{P^{out}}{P_{\max}^{out}} \right)^{\lambda_2 - 1} \quad (3)$$

Where the $PS(P^{out})$ represents the solar output PDF function dependent on the solar output value P^{out} , and $P_{\max}^{out}$ represents the maximum value of the P^{out} .

B. Wind Power:

The wind power output primarily depends upon the wind speed, and uncertainty in the wind speed needs to be modeled as a PDF. The literature demonstrates that the typical wind

speed is modeled as the Weibull PDF [18]. The expression of the wind speed as a Weibull PDF is shown below.

$$WS(v) = \left(\frac{s}{\alpha} \right) \left(\frac{v}{\alpha} \right)^{s-1} \exp \left[- \left(\frac{v}{\alpha} \right)^s \right] \quad (4)$$

Where v represents the wind speed; s and α are the shape and scale factors respectively. Now referring to [18], the relation between wind power and wind speed can be written as

$$P^W(v) = \begin{cases} 0 & v < v_{in}, v < v_{out} \\ \frac{v - v_{in}}{v_{rated} - v_{in}} P_{rated}^W & v_{in} \leq v \leq v_{rated} \\ P_{rated}^W & v_{rated} \leq v \leq v_{out} \end{cases} \quad (5)$$

Where P^W and P_{rated}^W represents the wind power and its rated value respectively. v_{in} and v_{out} are the cut-in and cut-out wind speeds, respectively. Referring to the equation (4) and (5), the expression for the wind power PDF can be written as

$$WP(P^W) = \begin{cases} \left(\frac{sr v_{in}}{\alpha P_{rated}^W} \right) \left[\left(\frac{(1 + r P^W / P_{rated}^W) v_{in}}{\alpha} \right)^s \right]^{s-1} \\ \times \exp \left\{ - \left[\left(\frac{(1 + r P^W / P_{rated}^W) v_{in}}{\alpha} \right)^s \right] \right\}, & P^W \in [0, P_{rated}^W] \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

Where $r = (v_{rated} / v_{in}) - 1$ is called the speed ratio.

C. EV arrival time at BSS:

In [19],[20], the EV arrival time at the BSS is modeled as Poisson distribution. The expression for the same is written as

$$\text{Prob}\{N_{ev}(t) = n_{ev}\} = \frac{e^{-\lambda_{ev}} (\lambda_{ev} t)^{n_{ev}}}{n_{ev}!}, \quad n_{ev} = 0, 1, \dots \quad (7)$$

Where the term $N_{ev}(t)$ represents the number of electric vehicles arriving at time t . Now λ_{ev} represents the vehicle arrival rate.

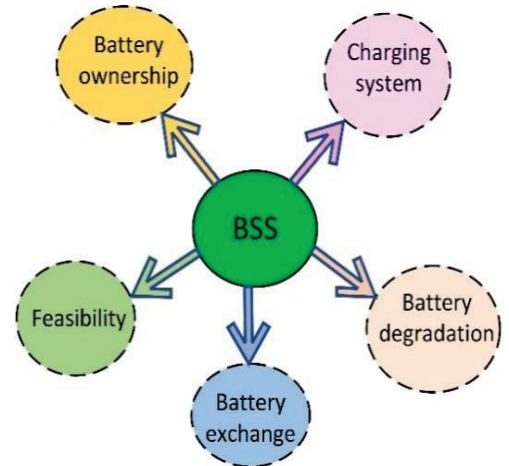


Fig. 3. Challenges associated with the BSS implementation.

VI. CHALLENGES

The significant problems limiting the BSS implementation are battery ownership, Interchangeability, BSS battery degradation, proper feasibility study, and suitable battery charging system inside the BSS. Even if many pilot projects are going on BSS infrastructure around the world, its cost of ownership is very high and is restricting the big players in the market from investing in it. The batteries used in the BSS need to be more robust in design, and very few battery manufacturers have that kind of robustness, which leads to less feasibility. Over a period, the BSS batteries will have degradation problems, and from the EV owner's perspective, they will always want a new battery pack should be replaced.

So, that will limit the maximum utilization of the batteries' operating cycle, which adds more cost. The battery charging system inside the BSS needs to have an advanced battery management system that will interact with the power system and provide/receive power during peak/off-peak hours. So, more cost is associated with it compared to CS. Hence more research needs to be done in this area to overcome these challenges to the BSS implementation. Fig. 3 shows the significant challenges associated with the BSS.

VII. CONCLUSION

This paper provides a comprehensive review of the battery swapping system and integration of BSS with the microgrid to provide additional support by acting as an energy storage device. We discussed most of the recent research done on these two areas. The discussed challenges associated with the BSS implementation will boost the readers to find the potential solution to mitigate them. We believe this paper will provide a comprehensive understating of the advantage of using BSS instead of conventional charging methods.

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